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MEMORY COMMAND VERIFICATION

Abstract

Methods, systems, and devices for performing memory command verification are described. A system may include a memory device and a memory controller, which may be external (e.g., a host device). The memory device may receive, from the memory controller, a command indicating a type of operation and an address. The memory device may decode the command and execute an operation (e.g., the operation corresponding to the decoded command) at an execution location on the memory device. The system (e.g., the memory device or the memory controller) may determine whether the executed operation and execution location match the type of operation and address indicated in the command, and the system may thereby determine an error associated with the decoding, the execution, or both of the command.

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Background/Summary

CROSS REFERENCE [0001] The present application for patent is a continuation of U.S. patent application Ser. No. 18/591,932 by Schaefer et al., entitled “MEMORY COMMAND VERIFICATION,” filed Feb. 29, 2024, which is a continuation of U.S. patent application Ser. No. 17/470,594 by Schaefer et al., entitled “MEMORY COMMAND VERIFICATION,” filed Sep. 9, 2021, which is a continuation of U.S. patent application Ser. No. 16/579,095 by Schaefer et al., entitled “MEMORY COMMAND VERIFICATION,” filed Sep. 23, 2019, which claims priority to U.S. Provisional Patent Application No. 62/746,274 by Schaefer et al., entitled “MEMORY COMMAND VERIFICATION,” filed Oct. 16, 2018, each of which is assigned to the assignee hereof and is expressly incorporated by reference in its entirety herein.

BACKGROUND

[0002] The following relates generally to a system that includes at least one memory device and more specifically to memory command verification.

[0003] Memory devices are widely used to store information in various electronic devices such as computers, wireless communication devices, cameras, digital displays, and the like. Information is stored by programming different states of a memory device. For example, binary devices most often store one of two states, often denoted by a logic 1 or a logic 0. In other devices, more than two states may be stored. To access the stored information, a component of the device may read, or sense, at least one stored state in the memory device. To store information, a component of the device may write, or program, the state in the memory device.

[0004] Types of memory devices include magnetic hard disks, random access memory (RAM), read-only memory (ROM), dynamic RAM (DRAM), synchronous dynamic RAM (SDRAM), ferroelectric RAM (FeRAM), magnetic RAM (MRAM), resistive RAM (RRAM), flash memory, phase change memory (PCM), and others. Memory devices may be volatile or non-volatile. Non-volatile memory, e.g., FeRAM, may maintain their stored logic state for extended periods of time even in the absence of an external power source. Volatile memory devices, e.g., DRAM, SRAM, may lose their stored state when disconnected from an external power source. Dynamic memory devices, e.g., DRAM, SDRAM, may lose a stored state over time unless they are periodically refreshed.

[0005] In some cases, an external memory controller may communicate commands to a memory device. The command may indicate a type of instruction to be executed at a given location on the memory device. Improving reliability for executing commands at a memory device may be desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 illustrates an example of a system that supports memory command verification as

disclosed herein.

[0007] FIG. 2 illustrates an example of a memory sub-array that supports memory command verification as disclosed herein.

[0008] FIGS. 3 and 4 illustrate examples of systems that support memory command verification as disclosed herein.

[0009] FIGS. 5 through 7 illustrate examples of timing diagrams that support memory command verification as disclosed herein.

[0010] FIGS. 8 and 9 illustrate block diagrams for apparatuses that support memory command verification as disclosed herein.

[0011] FIGS. 10 through 13 illustrate a method or methods that support memory command verification as disclosed herein.

DETAILED DESCRIPTION

[0012] Memory devices may operate under various conditions as part of electronic apparatuses such as personal computers, wireless communication devices, servers, internet-of-things (IoT) devices, electronic components of automotive vehicles, and the like. In some cases, memory devices supporting applications for certain implementations (e.g., automotive vehicles, in some cases with autonomous or semi-autonomous driving capabilities) may be subject to increased reliability constraints. As such, memory devices (e.g., DRAM) for some applications may be expected to operate with a reliability subject to relatively higher industry specifications (e.g., higher reliability constraints).

[0013] Some memory systems may include a host device (e.g., an external memory controller) which may transmit access commands (e.g., for a read operation, a write operation, a refresh operation) to a memory device. The memory device may receive the command from the host device and decode the command to determine an access instruction and an address (e.g., the access instruction indicating a type of access operation to be performed at the address of the memory device). The memory device may execute the access instruction at an execution address based on the decoded command. In some systems, the host device may not be able to determine errors associated with decoding and/or executing the command. For example, the memory device may determine (and execute) a type of access operation that is different than the type of access operation indicated by the command. Additionally or alternatively, the memory device may determine and/or execute the access operation at an address that is different than the address indicated by the command. A failure to detect such errors by the host device may negatively affect the reliability of the system.

[0014] Techniques for memory command verification are described. For example, the system may compare the access instruction and address included in the command to the executed access instruction and execution address to determine errors associated with the decoding and/or the execution of the command (e.g., whether any such error occurs). Determining errors associated with a specific command may allow the host device to retransmit commands not executed properly, thus increasing the reliability of the system. In some cases, the memory device may compare the executed access instruction and execution address to the command and transmit an indication relating to the comparison to the host device. In some other cases, the host device may compare the executed access instruction and execution address to the command.

[0015] Features of the disclosure are initially described in the context of memory systems and a memory device with reference to FIGS. 1-4. Features of the disclosure are described in the context of process flows with reference to FIGS. 5-7. These and other features of the disclosure are further illustrated by and described with reference to apparatus diagrams and flowcharts in FIGS. 8-13 that relate to memory command verification.

[0016] FIG. 1 illustrates an example of a system 100 that utilizes one or more memory devices in accordance with aspects disclosed herein. The system 100 may include an external memory controller 105, a memory device 110, and a plurality of channels 115 coupling the external memory

controller **105** with the memory device **110**. The system **100** may include one or more memory devices, but for ease of description the one or more memory devices may be described as a single memory device **110**.

[0017] The system **100** may include aspects of an electronic device, such as a computing device, a mobile computing device, a wireless device, or a graphics processing device. The system **100** may be an example of a portable electronic device. The system **100** may be an example of a computer, a laptop computer, a tablet computer, a smartphone, a cellular phone, a wearable device, an internet-connected device, or the like. The memory device **110** may be component of the system configured to store data for one or more other components of the system **100**. In some examples, the system **100** is configured for bi-directional wireless communication with other systems or devices using a base station or access point. In some examples, the system **100** is capable of machine-type communication (MTC), machine-to-machine (M2M) communication, or device-to-device (D2D) communication.

[0018] At least portions of the system **100** may be examples of a host device. Such a host device may be an example of a device that uses memory to execute processes such as a computing device, a mobile computing device, a wireless device, a graphics processing device, a computer, a laptop computer, a tablet computer, a smartphone, a cellular phone, a wearable device, an internet-connected device, some other stationary or portable electronic device, or the like. In some examples, system **100** is a graphics card. In some cases, the host device may refer to the hardware, firmware, software, or a combination thereof that implements the functions of the external memory controller **105**. In some cases, the external memory controller **105** may be referred to as a host or host device. The external memory controller **105** may communicate access commands to a memory device **110** indicating a type of operation (e.g., a read operation, a refresh operation, a write operation) to be executed at a memory array **170** at a location indicated by an address of the command. In some cases, the external memory controller **105** may be configured to detect errors associated with decoding and/or executing the access command at the memory device **110**. In some examples, the memory device **110** may indicate the error to the external memory controller **105**. In some other examples, the memory device **110** may transmit information related to the decoding and/or execution of the access command to enable the external memory controller **105** to detect the error.

[0019] In some cases, the memory device **110** may be an independent device or component that is configured to be in communication with other components of the system **100** and provide physical memory addresses/space to potentially be used or referenced by the system **100**. In some examples, a memory device **110** may be configurable to work with at least one or a plurality of different types of systems **100**. Signaling between the components of the system **100** and the memory device **110** may be operable to support modulation schemes to modulate the signals, different pin designs for communicating the signals, distinct packaging of the system **100** and the memory device **110**, clock signaling and synchronization between the system **100** and the memory device **110**, timing conventions, and/or other factors.

[0020] The memory device **110** may be configured to store data for the components of the system **100**. In some cases, the memory device **110** may act as a slave-type device to the system **100** (e.g., responding to and executing commands provided by the system **100** through the external memory controller **105**). Such commands may include an access command for an access operation, such as a write command for a write operation, a read command for a read operation, a refresh command for a refresh operation, or other commands. The memory device **110** may include two or more memory dice **160** (e.g., memory chips) to support a desired or specified capacity for data storage. The memory device **110** including two or more memory dice may be referred to as a multi-die memory or package (also referred to as multi-chip memory or package).

[0021] The system **100** may further include a processor **120**, a basic input/output system (BIOS) component **125**, one or more peripheral components **130**, and an input/output (I/O) controller **135**.

The components of system **100** may be in electronic communication with one another using a bus **140**.

[0022] The processor **120** may be configured to control at least portions of the system **100**. The processor **120** may be a general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or it may be a combination of these types of components. In such cases, the processor **120** may be an example of a central processing unit (CPU), a graphics processing unit (GPU), a general purpose GPU (GPGPU), or a system on a chip (SoC), among other examples.

[0023] The BIOS component **125** may be a software component that includes a BIOS operated as firmware, which may initialize and run various hardware components of the system **100**. The BIOS component **125** may also manage data flow between the processor **120** and the various components of the system **100**, e.g., the peripheral components **130**, the I/O controller **135**, etc. The BIOS component **125** may include a program or software stored in read-only memory (ROM), flash memory, or any other non-volatile memory.

[0024] The peripheral component(s) **130** may be any input device or output device, or an interface for such devices, that may be integrated into or with the system **100**. Examples may include disk controllers, sound controller, graphics controller, Ethernet controller, modem, universal serial bus (USB) controller, a serial or parallel port, or peripheral card slots, such as peripheral component interconnect (PCI) or accelerated graphics port (AGP) slots. The peripheral component(s) **130** may be other components understood by those skilled in the art as peripherals.

[0025] The I/O controller **135** may manage data communication between the processor **120** and the peripheral component(s) **130**, input devices **145**, or output devices **150**. The I/O controller **135** may manage peripherals that are not integrated into or with the system **100**. In some cases, the I/O controller **135** may represent a physical connection or port to external peripheral components.

[0026] The input **145** may represent a device or signal external to the system **100** that provides information, signals, or data to the system **100** or its components. This may include a user interface or interface with or between other devices. In some cases, the input **145** may be a peripheral that interfaces with system **100** via one or more peripheral components **130** or may be managed by the I/O controller **135**.

[0027] The output **150** may represent a device or signal external to the system **100** configured to receive an output from the system **100** or any of its components. Examples of the output **150** may include a display, audio speakers, a printing device, or another processor on printed circuit board, and so forth. In some cases, the output **150** may be a peripheral that interfaces with the system **100** via one or more peripheral components **130** or may be managed by the I/O controller **135**.

[0028] The components of system **100** may be made up of general-purpose or special purpose circuitry designed to carry out their functions. This may include various circuit elements, for example, conductive lines, transistors, capacitors, inductors, resistors, amplifiers, or other active or passive elements, configured to carry out the functions described herein.

[0029] The memory device **110** may include a device memory controller **155** and one or more memory dice **160**. Each memory die **160** may include a local memory controller **165** (e.g., local memory controller **165-a**, local memory controller **165-b**, and/or local memory controller **165-N**) and a memory array **170** (e.g., memory array **170-a**, memory array **170-b**, and/or memory array **170-N**). A memory array **170** may be a collection (e.g., a grid) of memory cells, with each memory cell being configured to store at least one bit of digital data. One or more memory arrays **170** may include a row access logic component and a column access logic component. The row and column access logic components may determine a subset of the memory array **170** accessed during an access operation indicated by the external memory controller **105**. In some examples, the subset of the memory array **170** may indicate the subset along one or more dimensions of the array. The memory array **170** may indicate the subset to a device memory controller **155**, a local memory

controller **165**, the external memory controller **105**, or a verifier component (e.g., as discussed in more detail with reference to FIGS. **3** and **4**). The system may compare the subset to an address indicated by the access operation in order to detect errors associated with decoding and/or executing the access command. Features of memory arrays **170** and/or memory cells are described in more detail with reference to FIG. **2**.

[0030] The memory device **110** may be an example of a two-dimensional (2D) array of memory cells or may be an example of a three-dimensional (3D) array of memory cells. For example, a 2D memory device may include a single memory die **160**. A 3D memory device may include two or more memory dice **160** (e.g., memory die **160-a**, memory die **160-b**, and/or any number of memory dice **160-N**). In a 3D memory device, a plurality of memory dice **160-N** may be stacked on top of one another. In some cases, memory dice **160-N** in a 3D memory device may be referred to as decks, levels, layers, or dies. A 3D memory device may include any quantity of stacked memory dice **160-N** (e.g., two high, three high, four high, five high, six high, seven high, eight high). This may increase the number of memory cells that may be positioned on a substrate as compared with a single 2D memory device, which in turn may reduce production costs or increase the performance of the memory array, or both. In some 3D memory device, different decks may share at least one common access line such that some decks may share at least one of a word line, a digit line, and/or a plate line.

[0031] The device memory controller **155** may include circuits or components configured to control operation of the memory device **110**. As such, the device memory controller **155** may include the hardware, firmware, and software that enables the memory device **110** to perform commands and may be configured to receive, transmit, or execute commands, data, or control information related to the memory device **110**. The device memory controller **155** may be configured to communicate with the external memory controller **105**, the one or more memory dice **160**, or the processor **120**. In some cases, the memory device **110** may receive data and/or commands from the external memory controller **105**. For example, the memory device **110** may receive a write command indicating that the memory device **110** is to store certain data on behalf of a component of the system **100** (e.g., the processor **120**) or a read command indicating that the memory device **110** is to provide certain data stored in a memory die **160** to a component of the system **100** (e.g., the processor **120**). In some cases, the device memory controller **155** may control operation of the memory device **110** described herein in conjunction with the local memory controller **165** of the memory die **160**. Examples of the components included in the device memory controller **155** and/or the local memory controllers **165** may include receivers for demodulating signals received from the external memory controller **105**, decoders for modulating and transmitting signals to the external memory controller **105**, logic, decoders, amplifiers, filters, or the like.

[0032] The local memory controller **165** (e.g., local to a memory die **160**) may be configured to control operations of the memory die **160**. Also, the local memory controller **165** may be configured to communicate (e.g., receive and transmit data and/or commands) with the device memory controller **155**. The local memory controller **165** may support the device memory controller **155** to control operation of the memory device **110** as described herein. In some cases, the memory device **110** does not include the device memory controller **155**, and the local memory controller **165** or the external memory controller **105** may perform the various functions described herein. As such, the local memory controller **165** may be configured to communicate with the device memory controller **155**, with other local memory controllers **165**, or directly with the external memory controller **105** or the processor **120**.

[0033] The external memory controller **105** may be configured to enable communication of information, data, and/or commands between components of the system **100** (e.g., the processor **120**) and the memory device **110**. The external memory controller **105** may act as a liaison between the components of the system **100** and the memory device **110** so that the components of the

system **100** may not need to know the details of the memory device's operation. The components of the system **100** may present requests to the external memory controller **105** (e.g., read commands or write commands) that the external memory controller **105** satisfies. The external memory controller **105** may convert or translate communications exchanged between the components of the system **100** and the memory device **110**. In some cases, the external memory controller **105** may include a system clock that generates a common (source) system clock signal. In some cases, the external memory controller **105** may include a common data clock that generates a common (source) data clock signal.

[0034] In some cases, the external memory controller **105** or other component of the system **100**, or its functions described herein, may be implemented by the processor **120**. For example, the external memory controller **105** may be hardware, firmware, or software, or some combination thereof implemented by the processor **120** or other component of the system **100**. While the external memory controller **105** is depicted as being external to the memory device **110**, in some cases, the external memory controller **105**, or its functions described herein, may be implemented by a memory device **110**. For example, the external memory controller **105** may be hardware, firmware, or software, or some combination thereof implemented by the device memory controller **155** or one or more local memory controllers **165**. In some cases, the external memory controller **105** may be distributed across the processor **120** and the memory device **110** such that portions of the external memory controller **105** are implemented by the processor **120** and other portions are implemented by a device memory controller **155** or a local memory controller **165**. Likewise, in some cases, one or more functions ascribed herein to the device memory controller **155** or local memory controller **165** may in some cases be performed by the external memory controller **105** (either separate from or as included in the processor **120**).

[0035] The components of the system **100** may exchange information with the memory device **110** using a plurality of channels **115**. In some examples, the channels **115** may enable communications between the external memory controller **105** and the memory device **110**. Each channel **115** may include one or more signal paths or transmission mediums (e.g., conductors) between terminals associated with the components of system **100**. For example, a channel **115** may include a first terminal including one or more pins or pads at external memory controller **105** and one or more pins or pads at the memory device **110**. A pin may be an example of a conductive input or output point of a device of the system **100**, and a pin may be configured to act as part of a channel. In some cases, a pin or pad of a terminal may be part of to a signal path of the channel **115**. Additional signal paths may be coupled with a terminal of a channel for routing signals within a component of the system **100**. For example, the memory device **110** may include signal paths (e.g., signal paths internal to the memory device **110** or its components, such as internal to a memory die **160**) that route a signal from a terminal of a channel **115** to the various components of the memory device **110** (e.g., a device memory controller **155**, memory dice **160**, local memory controllers **165**, memory arrays **170**).

[0036] Channels **115** (and associated signal paths and terminals) may be dedicated to communicating specific types of information. In some cases, a channel **115** may be an aggregated channel and thus may include multiple individual channels. For example, a data channel **190** may be x4 (e.g., including four signal paths), x8 (e.g., including eight signal paths), x16 (including sixteen signal paths), and so forth.

[0037] In some cases, the channels **115** may include one or more command and address (CA) channels **186**. The CA channels **186** may be configured to communicate commands between the external memory controller **105** and the memory device **110** including control information associated with the commands (e.g., address information). For example, the CA channel **186** may include a read command with an address of the desired data. In some cases, the CA channels **186** may be registered on a rising clock signal edge and/or a falling clock signal edge. In some cases, a CA channel **186** may include eight or nine signal paths.

[0038] In some cases, the channels **115** may include one or more clock signal (CK) channels **188**. The CK channels **188** may be configured to communicate one or more common clock signals between the external memory controller **105** and the memory device **110**. Each clock signal may be configured to oscillate between a high state and a low state and coordinate the actions of the external memory controller **105** and the memory device **110**. In some cases, the clock signal may be a differential output (e.g., a CK_t signal and a CK_c signal) and the signal paths of the CK channels **188** may be configured accordingly. In some cases, the clock signal may be single ended. In some cases, the clock signal may be a 1.5 GHz signal. A CK channel **188** may include any number of signal paths. In some cases, the clock signal CK (e.g., a CK_t signal and a CK_c signal) may provide a timing reference for command and addressing operations for the memory device **110**, or other system-wide operations for the memory device **110**. The clock signal CK therefore may be variously referred to as a control clock signal CK, a command clock signal CK, or a system clock signal CK. The system clock signal CK may be generated by a system clock, which may include one or more hardware components (e.g., oscillators, crystals, logic gates, transistors, or the like).

[0039] In some cases, the channels **115** may include one or more data (DQ) channels **190**. The data channels **190** may be configured to communicate data and/or control information between the external memory controller **105** and the memory device **110**. For example, the data channels **190** may communicate information (e.g., bi-directional) to be written to the memory device **110** or information read from the memory device **110**. The data channels **190** may communicate signals that may be modulated using a variety of different modulation schemes (e.g., NRZ, PAM4).

[0040] In some cases, the channels **115** may include one or more other channels **192** that may be dedicated to other purposes. These other channels **192** may include any number of signal paths.

[0041] In some cases, the other channels **192** may include one or more write clock signal (WCK) channels. While the 'W' in WCK may nominally stand for "write," a write clock signal WCK (e.g., a WCK_t signal and a WCK_c signal) may provide a timing reference for access operations generally for the memory device **110** (e.g., a timing reference for both read and write operations). Accordingly, the write clock signal WCK may also be referred to as a data clock signal WCK. The WCK channels may be configured to communicate a common data clock signal between the external memory controller **105** and the memory device **110**. The data clock signal may be configured to coordinate an access operation (e.g., a write operation or read operation) of the external memory controller **105** and the memory device **110**. In some cases, the write clock signal may be a differential output (e.g., a WCK_t signal and a WCK_c signal) and the signal paths of the WCK channels may be configured accordingly. A WCK channel may include any number of signal paths. The data clock signal WCK may be generated by a data clock, which may include one or more hardware components (e.g., oscillators, crystals, logic gates, transistors, or the like).

[0042] In some cases, the other channels **192** may include one or more error detection code (EDC) channels. The EDC channels may be configured to communicate error detection signals, such as checksums, to improve system reliability. An EDC channel may include any number of signal paths.

[0043] The channels **115** may couple the external memory controller **105** with the memory device **110** using a variety of different architectures. Examples of the various architectures may include a bus, a point-to-point connection, a crossbar, a high-density interposer such as a silicon interposer, or channels formed in an organic substrate or some combination thereof. For example, in some cases, the signal paths may at least partially include a high-density interposer, such as a silicon interposer or a glass interposer.

[0044] Signals communicated over the channels **115** may be modulated using a variety of different modulation schemes. In some cases, a binary-symbol (or binary-level) modulation scheme may be used to modulate signals communicated between the external memory controller **105** and the memory device **110**. A binary-symbol modulation scheme may be an example of an M-ary

modulation scheme where M is equal to two. Each symbol of a binary-symbol modulation scheme may be configured to represent one bit of digital data (e.g., a symbol may represent a logic 1 or a logic 0). Examples of binary-symbol modulation schemes include, but are not limited to, non-return-to-zero (NRZ), unipolar encoding, bipolar encoding, Manchester encoding, pulse amplitude modulation (PAM) having two symbols (e.g., PAM2), and/or others.

[0045] In some cases, a multi-symbol (or multi-level) modulation scheme may be used to modulate signals communicated between the external memory controller **105** and the memory device **110**. A multi-symbol modulation scheme may be an example of an M-ary modulation scheme where M is greater than or equal to three. Each symbol of a multi-symbol modulation scheme may be configured to represent more than one bit of digital data (e.g., a symbol may represent a logic 00, a logic 01, a logic 10, or a logic 11). Examples of multi-symbol modulation schemes include, but are not limited to, PAM4, PAM8, etc., quadrature amplitude modulation (QAM), quadrature phase shift keying (QPSK), and/or others. A multi-symbol signal or a PAM4 signal may be a signal that is modulated using a modulation scheme that includes at least three levels to encode more than one bit of information. Multi-symbol modulation schemes and symbols may alternatively be referred to as non-binary, multi-bit, or higher-order modulation schemes and symbols.

[0046] FIG. 2 illustrates an example of a memory sub-array **200** in accordance with various examples of the present disclosure. The memory sub-array **200** may be an example of at least a portion of the memory dice **160** described with reference to FIG. 1. In some cases, the memory sub-array **200** may be referred to as a memory die, memory chip, a memory device, or an electronic memory apparatus. For example, a memory device such as a memory chip may include multiple instances of sub-array **200**, with additional row, address, bank, or bank group decoding used to select one or more sub-arrays from the multiple instances for access operations. The memory sub-array **200** may include one or more memory cells **205** that are programmable to store different logic states. Each memory cell **205** may be programmable to store two or more states. For example, the memory cell **205** may be configured to store one bit of digital logic at a time (e.g., a logic 0 and a logic 1). In some cases, a single memory cell **205** (e.g., a multi-level memory cell) may be configured to store more than one bit of digit logic at a time (e.g., a logic 00, logic 01, logic 10, or a logic 11).

[0047] A memory cell **205** may store a charge representative of the programmable states in a capacitor. DRAM architectures may include a capacitor that includes a dielectric material to store a charge representative of the programmable state. In other memory architectures, other storage devices and components are possible. For example, nonlinear dielectric materials may be employed.

[0048] Operations such as reading and writing may be performed on memory cells **205** by activating or selecting access lines such as a word line **210** and/or a digit line **215**. In some cases, digit lines **215** may also be referred to as bit lines. References to access lines, word lines and digit lines, or their analogues, are interchangeable without loss of understanding or operation. Activating or selecting a word line **210** or a digit line **215** may include applying a voltage to the respective line or configuring a multiplexer to map the line to a given signal.

[0049] The memory sub-array **200** may include the access lines (e.g., the word lines **210** and the digit lines **215**) arranged in a grid-like pattern. Memory cells **205** may be positioned at intersections of the word lines **210** and the digit lines **215**. By biasing a word line **210** (e.g., applying a voltage to the word line **210**), a memory cell **205** may be accessed via the digit line **215** at their intersection.

[0050] Accessing the memory cells **205** may be controlled through a row decoder **220** or a column decoder **225**. For example, a row decoder **220** may receive a row address from the local memory controller **260** and activate a word line **210** based on the received row address. A column decoder **225** may receive a column address from the local memory controller **260** and may select a digit line **215** based on the received column address. For example, the memory sub-array **200** may include multiple word lines **210**, labeled WL_1 through WL_M, and multiple digit lines **215**, labeled DL_1

through DL_N, where M and N depend on the size of the memory array. Thus, by activating a word line **210**, e.g., WL₁, the memory cells **205** in a given row may be accessed. The digit lines **215** (e.g., DL₁, . . . , DL_N) carry the data for writing or reading from the memory cells in the row. The intersection of a word line **210** and a digit line **215**, in either a two-dimensional or three-dimensional configuration, may be referred to as an address of a memory cell **205**.

[0051] The memory cell **205** may include a logic storage component, such as capacitor **230** and a switching component **235**. The capacitor **230** may be an example of a dielectric capacitor or a ferroelectric capacitor. A first node of the capacitor **230** may be coupled with the switching component **235** and a second node of the capacitor **230** may be coupled with a voltage source **240**. In some cases, the voltage source **240** may be the cell plate reference voltage, such as V_{pl}, or may be ground, such as V_{ss}. In some cases, the voltage source **240** may be an example of a plate line coupled with a plate line driver. The switching component **235** may be an example of a transistor or any other type of switch device that selectively establishes or de-establishes electronic communication between two components.

[0052] Selecting or deselecting the memory cell **205** may be accomplished by activating or deactivating the switching component **235**. The capacitor **230** may be in electronic communication with the digit line **215** using the switching component **235**. For example, the capacitor **230** may be isolated from digit line **215** when the switching component **235** is deactivated, and the capacitor **230** may be coupled with digit line **215** when the switching component **235** is activated. In some cases, the switching component **235** is a transistor and its operation may be controlled by applying a voltage to the transistor gate, where the voltage differential between the transistor gate and transistor source may be greater or less than a threshold voltage of the transistor. In some cases, the switching component **235** may be a p-type transistor or an n-type transistor. The word line **210** may be in electronic communication with the gate of the switching component **235** and may activate/deactivate the switching component **235** based on a voltage being applied to word line **210**.

[0053] A word line **210** may be a conductive line in electronic communication with a memory cell **205** that is used to perform access operations on the memory cell **205**. In some architectures, the word line **210** may be in electronic communication with a gate of a switching component **235** of a memory cell **205** and may be configured to control the switching component **235** of the memory cell. In some architectures, the word line **210** may be in electronic communication with a node of the capacitor of the memory cell **205** and the memory cell **205** may not include a switching component.

[0054] The memory sub-array **200** may include row access logic **265**. The row access logic **265** may be coupled to the word lines **210** or to portions of row decoder **220**. The row access logic **265** may be configured to determine a word line **210** that is activated during an access operation. In some cases, the row access logic **265** may determine a single word line **210** that is activated during the access operation. In some other region, the row access logic **265** may determine a subset of word lines **210** including the word line that is activated during the access operation. The row access logic **265** may be configured to transmit an indication of the word line or lines **210** that are activated during the access operation to a component (e.g., a verifier as described herein **235**).

[0055] A digit line **215** may be a conductive line that connects the memory cell **205** with a sense component **245**. In some architectures, the memory cell **205** may be selectively coupled with the digit line **215** during portions of an access operation. For example, the word line **210** and the switching component **235** of the memory cell **205** may be configured to couple and/or isolate the capacitor **230** of the memory cell **205** and the digit line **215**. In some architectures, the memory cell **205** may be in electronic communication (e.g., constant) with the digit line **215**. The memory sub-array **200** may include column access logic **275**. The column access logic **275** may be coupled to the column decoder **225** or the sense component **245**. The column access logic **275** may be configured to determine a digit line **215** that is selected during an access operation. In some cases, the column access logic **275** may determine a single digit line **215** that is selected during the access

operation. In some other cases, the column access logic 275 may determine a subset of digit lines 215 that are selected (or that include one or more digit lines that are selected) during the access operation. The column access logic 275 may be configured to transmit an indication of the digit line or lines 215 that are selected during the access operation to a component (e.g., a verifier as described herein).

[0056] The sense component 245 may be configured to detect a state (e.g., a charge) stored on the capacitor 230 of the memory cell 205 and determine a logic state of the memory cell 205 based on the stored state. The charge stored by a memory cell 205 may be extremely small, in some cases. As such, the sense component 245 may include one or more sense amplifiers to amplify the signal output by the memory cell 205. The sense amplifiers may detect small changes in the charge of a digit line 215 during a read operation and may produce signals corresponding to a logic state 0 or a logic state 1 based on the detected charge. During a read operation, the capacitor 230 of memory cell 205 may output a signal (e.g., via charge sharing) to its corresponding digit line 215. The signal may cause a voltage of the digit line 215 to change. The sense component 245 may be configured to compare the signal received from the memory cell 205 across the digit line 215 to a reference signal 250 (e.g., reference voltage). The sense component 245 may determine the stored state of the memory cell 205 based on the comparison. For example, in binary-signaling, if digit line 215 has a higher voltage than the reference signal 250, the sense component 245 may determine that the stored state of memory cell 205 is a logic 1 and, if the digit line 215 has a lower voltage than the reference signal 250, the sense component 245 may determine that the stored state of the memory cell 205 is a logic 0. The sense component 245 may include amplifiers (e.g., transistor amplifiers) to detect and amplify a difference in the signals. The detected logic state of memory cell 205 may be output through column decoder 225 as output 255. In some cases, the aspects of sense component 245 may be part of another component (e.g., a column decoder 225, row decoder 220). In some cases, the sense component 245 may be in electronic communication with the row decoder 220 or the column decoder 225.

[0057] The local memory controller 260 may control the operation of memory cells 205 through the various components (e.g., row decoder 220, column decoder 225, and sense component 245). The local memory controller 260 may be an example of the local memory controller 165 described with reference to FIG. 1. In some cases, aspects of the row decoder 220, column decoder 225, or sense component 245 may be co-located with the local memory controller 260. The local memory controller 260 may be configured to receive commands and/or data from an external memory controller 105 (or a device memory controller 155 described with reference to FIG. 1), translate the commands and/or data into information that can be used by the memory sub-array 200, perform one or more operations on the memory sub-array 200, and communicate data from the memory sub-array 200 to the external memory controller 105 (or the device memory controller 155) in response to performing the one or more operations. The local memory controller 260 may generate row and column address signals to activate the target word line 210 and select the target digit line 215. The local memory controller 260 may also generate and control various voltages or currents used during the operation of the memory sub-array 200. In general, the amplitude, shape, or duration of an applied voltage or current discussed herein may be adjusted or varied and may be different for the various operations discussed in operating the memory sub-array 200.

[0058] In some cases, the local memory controller 260 may be configured to perform a write operation (e.g., a programming operation) on one or more memory cells 205 of the memory sub-array 200. During a write operation, a memory cell 205 of the memory sub-array 200 may be programmed to store a desired logic state. In some cases, a plurality of memory cells 205 may be programmed during a single write operation. The local memory controller 260 may identify a target memory cell 205 on which to perform the write operation. The local memory controller 260 may identify a target word line 210 and a target digit line 215 in electronic communication with the target memory cell 205 (e.g., the address of the target memory cell 205). The local memory

controller **260** may activate the target word line **210** (e.g., applying a voltage to the word line **210**), to access a row including the target memory cell **205**. The local memory controller **260** may apply a specific signal (e.g., voltage) to the digit line **215** during the write operation to store a specific state (e.g., charge) in the capacitor **230** of the memory cell **205**, the specific state (e.g., charge) may be indicative of a desired logic state.

[0059] The row access logic **265** and the column access logic **275** may determine a subset of the array corresponding to the target memory cell **205**. That is, the row access logic **265** and the column access logic **275** may be configured to determine the word line or lines **210** activated by the write operation and the digit line or lines **215** that convey data associated with the write operation. The row access logic **265** and the column access logic **275** may transmit an indication of the determined subset of the array (e.g., to a verifier as discussed herein). In some examples, the indication of the subset of the array may indicate the subset along one or more dimensions of the array. For example, the indication of the subset of the array may indicate the word line or lines **210** being activated during the write operation (e.g., a row matrix (MAT) indication). Additionally or alternatively, the indication of the subset of the array may indicate the digit line or lines **215** selected during the write operation (e.g., a column MAT indication).

[0060] In some cases, the local memory controller **260** may be configured to perform a read operation (e.g., a sense operation) on one or more memory cells **205** of the memory sub-array **200**. During a read operation, the logic state stored in a memory cell **205** of the memory sub-array **200** may be determined. In some cases, a plurality of memory cells **205** may be sensed during a single read operation. The local memory controller **260** may identify a target memory cell **205** on which to perform the read operation. The local memory controller **260** may identify a target word line **210** and a target digit line **215** in electronic communication with the target memory cell **205** (e.g., the address of the target memory cell **205**). The local memory controller **260** may activate the target word line **210** (e.g., applying a voltage to the word line **210**), to access a row including the target memory cell **205**. The target memory cell **205** may transfer a signal to the sense component **245** in response to biasing the access lines. The sense component **245** may amplify the signal. The local memory controller **260** may fire the sense component **245** (e.g., latch the sense component) and thereby compare the signal received from the memory cell **205** to the reference signal **250**. Based on that comparison, the sense component **245** may determine a logic state that is stored on the memory cell **205**. The local memory controller **260** may communicate the logic state stored on the memory cell **205** to the external memory controller **105** (or the device memory controller **155**) as part of the read operation (e.g., by selecting data read from digit lines **215** using column decoder **225**).

[0061] The row access logic **265** and the column access logic **275** may determine a subset of the array corresponding to the target memory cell **205**. That is, the row access logic **265** and the column access logic **275** may be configured to determine the word line or lines **210** activated during the read operation and the digit line or lines **215** carrying the data associated with the read operation. The row access logic **265** and the column access logic **275** may transmit an indication of the determined subset of the array (e.g., to a verifier as discussed herein). In some examples, the indication of the subset of the array may indicate the subset along one or more dimensions of the array. For example, the indication of the subset of the array may indicate the word line or lines **210** being activated during the read operation (e.g., a row MAT indication). Additionally or alternatively, the indication of the subset of the array may indicate the digit line or lines **215** selected during the read operation (e.g., a column MAT indication).

[0062] In some memory architectures, accessing the memory cell **205** may degrade or destroy the logic state stored in a memory cell **205**. For example, a read operation performed in DRAM architectures may partially or completely discharge the capacitor of the target memory cell. The local memory controller **260** may perform a re-write operation or a refresh operation to return the memory cell to its original logic state. The local memory controller **260** may re-write the logic state

to the target memory cell after a read operation. In some cases, the re-write operation may be considered part of the read operation. Additionally, activating a single access line, such as a word line **210**, may disturb the state stored in some memory cells in electronic communication with that access line. Thus, a re-write operation or refresh operation may be performed on one or more memory cells that may not have been accessed.

[0063] FIG. **3** illustrates an example of a system **300** that supports techniques for memory command verification for a memory device. The system **300** may include one or more components described herein with reference to FIGS. **1** and **2**. For example, the system **300** may include a host device **305**, which may be an example of the external memory controller **105** as described with reference to FIG. **1**; a memory device **310**, which may be an example of the memory device **110**, the memory dice **160**, or the memory sub-array **200** as described with reference to FIGS. **1** and **2**; and a memory array **370**, which may be an example of a memory array **170** as described with reference to FIG. **1** and may include aspects of memory sub-array **200** as described with reference to FIG. **2**. The memory device **310** may also include the memory pins **315**, an input buffer **325**, and a verifier **345**.

[0064] Host device **305** may send a command via a channel **320**, which may be an example of a channel **115** or CA channel **186** as discussed with reference to FIG. **1**. The command may include an instruction for an access operation (e.g., a read operation, a refresh operation, a write operation) at memory array **370** on the memory device **310**. The command may further include an address of the memory array **370** targeted by the access operation. The address may comprise a row address and a column address corresponding to a row and column(s) of the memory array **370**. The memory device **310** may utilize a verifier **345** in order to determine (e.g., identify, detect) errors associated with the transferring, decoding, or execution of the command received from the host device **305**. In some cases, determining the errors may improve the reliability of the system **300**. For example, the host device **305** may retransmit the command or commands for which the memory device **310** determined an error, or may otherwise process additional commands to recover from the error.

[0065] Upon receiving the command at the memory pins **315**, the memory device **310** may latch (e.g., capture) the command at the input buffer **325** via bus **330**. The command may comprise raw pin data detected at the memory pins **315** from the channel **320**. The memory device **310** may transfer the command from the input buffer **325** to the decoder **335** (e.g., via bus **340**), where the decoder **335** may attempt to decode the command into an access instruction and an address (e.g., the access instruction indicating a type of access operation to be performed at the address of the memory array **370**). In some cases, there may be an error when the memory device **310** transfers the command from the input buffer **325** and/or attempts to decode the command at decoder **335**. For example, the command may not be transferred correctly from the input buffer **325** to the decoder **335**. That is, the raw pin data (e.g., indicating the command) received at the decoder **335** may not match the data transmitted across channel **320** from the host device **305** (e.g., the data received at the input buffer **325**). In another example, the decoder **335** may incorrectly decode the command and therefore may not determine a correct access instruction and/or address.

[0066] Verifier **345** may be configured to detect such errors. The memory device **310** may transmit data corresponding to the access instruction (e.g., as determined based on decoding the command) from the decoder **335** to the verifier **345** (e.g., via a bus **380**). In some cases, the data may indicate a success (or lack thereof) of decoding the command to determine an access instruction. That is, the data may indicate an error resulting in the decoder **335** not determining a type of access instruction based on the command (e.g., not successfully decoding the command). For example, the decoder **335** may transmit a logic '0' on the bus **380** in the event that an access instruction was determined from the command. Conversely, the decoder **335** may transmit a logic '1' on the bus **380** in the event that an access instruction was not determined from the command (e.g., corresponding to a failed decoding by decoder **335**). In some other cases, the data may indicate the type of access

instruction determined by the decoder **335**. For example, the decoder **335** may transmit a signal (or multiple signals) to the verifier **345** that may map to (e.g., identify) the type of access instruction (e.g., a read instruction, a refresh instruction, a write instruction).

[0067] In the case where the decoder **335** transmits the type of access instruction to the verifier **345**, the verifier **345** may determine if the type of access instruction determined at the decoder **335** matches a type of access instruction indicated by the command stored at the input buffer **325**. Here, the verifier **345** may also receive the command stored at the input buffer **325** (e.g., corresponding to the raw pin data received at the memory pins **315**) via bus **350**. The verifier may decode, by a second decoder **355**, the command received from the input buffer **325**. In some cases, the decoder **355** may decode the command and determine a type of access instruction (e.g., in a similar manner as decoder **335**). The verifier **345** may compare the type of access instruction determined at the decoder **355** to the type of access instruction determined at decoder **335** (e.g., by comparator **360**). By this method, the verifier **345** may determine if there is an error associated with the access instruction determined at the decoder **335** (e.g., if the comparator **360** determines that the two access instruction types are different).

[0068] After decoding the command received from the input buffer **325**, the decoder **335** may subsequently indicate the instruction and the address to the memory array **370** (e.g., via the bus **385**). The memory array **370** may access the memory array **370** at the indicated address and according to the indicated instruction (e.g., an instruction corresponding to a read operation, a refresh operation, a write operation). For example, the memory array **370** may include a row decoder and a column decoder (e.g., as described with reference to FIG. 2) configured to receive an address for memory array **370** and activate a word line and select digit lines respectively based on the received address (e.g., as received from the decoder **335**). The memory array **370** may further include row access logic **365** and column access logic **375** configured to determine an accessed portion of the memory array **370** accessed during the execution of the indicated instruction. For example, as described with reference to FIG. 2, the row access logic **365** may determine a word line or a set of word lines accessed by the memory array **370**. The word line or word lines accessed may correspond to an accessed row MAT. Additionally or alternatively, the column access logic **375** may determine a digit line or a set of digit lines accessed by the memory array **370**. The digit line or digit lines accessed may correspond to an accessed column MAT. The accessed row MAT and/or the access column MAT may correspond to the accessed portion of the memory array **370**.

[0069] There may be an error corresponding to the location of the memory array **370** during the access operation. For example, the decoder **335** may incorrectly decode the command and determine an address for the access command that does not correspond to an address indicated by the command received from the host device **305**. In another example, the memory array **370** may incorrectly access a word line and/or digit line. For example, a row decoder and/or a column decoder may activate an incorrect word or select an incorrect digit line (e.g., a different word or digit line than indicated by the address within the command received from the host device **305**).

[0070] Verifier **345** may be configured to detect such errors (e.g., via bus **390**). The memory array **370** may transmit an indication of the accessed portion (e.g., as determined by the row access logic **365** and/or the column access logic **375**) to the verifier **345**. The indication may include an identifier for the accessed portion (e.g., a row identifier, a column identifier, a row MAT identifier, a column MAT identifier). A row MAT may refer to a single word line, a subset of word lines of one or more sub-arrays, a sub-array, or the like. A column MAT may refer to a single digit line, a subset of digit lines of one or more sub-arrays, a sub-array, or the like.

[0071] The verifier **345** may receive the command from the input buffer **325**. As discussed above, the verifier **345** may utilize a second decoder **355**. In some cases, the second decoder **355** may be different than decoder **335**. For example, the decoder **335** may be optimized for decoding speed to support timing of various memory operations, while the decoder **355** may be optimized for robustness. In some cases, decoder **335** may be a custom logic or transistor level circuit, while

decoder 355 may be a synthesized logic block (e.g., synthesized from a hardware description language). In some cases, the decoder 335 may use switched logic gates while, decoder 355 uses static combinatorial logic gates. In a first example, the decoder 355 may decode the command received from the input buffer 325 to determine an instruction and an expected address of the memory array 370. In this example, the verifier 345 may compare the expected address determined at the second decoder 355 to the accessed portion indicated by the memory array 370 (e.g., at comparator 360). The verifier 345 may determine an error in the event that the accessed portion does not match a portion of the memory array 370 indicated by the expected address. Here, the memory device 310 may determine that the executed command (e.g., executed according to the address determined by the first decoder 335 which may be different than the address determined by the second decoder 355 and/or row or column decoders in memory array 370) may not have been executed properly.

[0072] In a second example, the decoder 355 may map the command received from the input buffer 325 to determine an expected portion of the memory array 370 containing the address indicated by the command (e.g., without determining the decoded address itself). That is, the portions of the memory array 370 (e.g., row MAT, column MAT) may be related to a physical structure of the memory array 370 and may not directly correspond to the decoded address. Thus, the decoded address may correspond to a logical or virtual address space, while the accessed portion of the memory array 370 may be represented in a physical address space. Here, the decoder 355 may map the command to an expected portion of the memory array 370. The expected portion may include an expected row MAT, an expected column MAT, or an expected array (e.g., an expected column MAT and an expected row MAT). In this example, the verifier 345 may compare the expected portion determined at the second decoder 355 to the accessed portion indicated by the memory array 370 (e.g., at comparator 360). The verifier 345 may determine an error in the event that the expected portion determined by the second decoder 355 does not correspond to the accessed portion indicated by the memory array 370. Here, the memory device 310 may determine that the executed command may not have been executed properly.

[0073] In the event that memory device 310 detects an error corresponding to the instruction and/or the location accessed by the instruction, the verifier 345 may indicate the error to the host device 305 (e.g., via bus 395). In some cases, in the event the memory device determines an error (such as incorrectly decoding the command, failing to decode the command, accessing a portion of the memory array 370 that does not match a portion of the memory array 370 indicated by the command), the memory device 310 may indicate the error at the sideband pins 396 (e.g., by activating a pin by the verifier 345 via bus 395). For example, raising the voltage of the pin to a logic '1' may indicate an error occurred during the execution of the received command while keeping the voltage of the pin at a ground voltage (e.g., a logic '0') may indicate that memory device 310 has not detected an error occurring during the execution of one or more commands. Alternatively, the logic states for the indication could be inverted from this example.

[0074] In another example, the verifier 345 may transmit additional data via bus 395, sideband pins 396, and bus 397. The sideband pins 396 may be a serial bus such that the verifier 345 sends the data serially via the sideband pins 396 and bus 397. The additional data may include the one or more of a success or failure for decoding the command by decoder 335, the access instruction type determined by the decoder 335, or the accessed portion of the memory array 370. In some cases, the additional data may allow the host device 305 to determine status of the execution of the command at the memory device 310. For example, the host device 305 may compare the access instruction type determined by the decoder 335 or the accessed portion of the memory array 370 to the intended instruction type or address. The host device 305 may process additional commands based on the status of execution of the command. For example, the host device 305 may determine to retransmit the command based on receiving an error indication from the memory device 310. This may allow the host device 305 to improve reliability of the memory device 310 when

compared to a memory device that does not provide such error feedback.

[0075] In some cases, the host device **305** may transmit a stream of commands to the memory device **310**. In this example, the memory device **310** may concurrently execute the transmitted commands as well as determine errors associated with the execution of one or more of the commands. That is, the communications and operations done at the verifier **345** may not disrupt the operations (e.g., normal memory access operations) of the memory device **310**. This may allow the memory device **310** to provide additional feedback to the host device **305** (e.g., increasing reliability of the memory device **310**) without decreasing (e.g., significantly or at all) read/write speeds of the memory device.

[0076] FIG. **4** illustrates an example of a system **400** that supports techniques for memory command verification for a memory device. The system **400** may include one or more components described herein with reference to FIGS. **1** and **2**. For example, the system **400** may include a host device **405**, which may be an example of the external memory controller **105** as described with reference to FIG. **1**; a memory device **410**, which may be an example of the memory device **110**, the memory dice **160**, or the memory sub-array **200** as described with reference to FIGS. **1** and **2**; and a memory array **470**, which may be an example of a memory array **170** as described with reference to FIG. **1** and may include aspects of memory sub-array **200** as described with reference to FIG. **2**. The memory device **410** may also include the memory pins **415**, and an input buffer **425**.

[0077] Host device **405** may send a command via a channel **420**, which may be an example of a channel **115** or CA channel **186** as discussed with reference to FIG. **1**. The command may include an instruction for an access operation (e.g., a read operation, a write operation) at a memory array **470** on the memory device **410**. The command may further include an address of the memory array **470** targeted by the access operation. The address may comprise a row address and a column address corresponding to a row and column at the memory array **470**. The memory device **410** may indicate, to the host device **405**, information related to a decoding of the command and/or information related to a portion of the memory array **470** accessed during an execution of the command. The indicated information may allow host device **405** to determine errors associated with transferring, decoding, and/or executing the command received from the host device **405**. In some cases, determining the errors may improve the reliability of the system **400**. For example, the host device **305** may retransmit the command or commands for which the host device **405** determined an error, or may otherwise process additional commands to recover from the error.

[0078] Upon receiving the command at the memory pins **415**, the memory device **410** may latch the command at the input buffer **425** via bus **430**. The command may comprise raw pin data detected at the memory pins **415** from the bus **420**. The memory device **410** may transfer the command from the input buffer **425** to the decoder **435** (e.g., via bus **440**), where the decoder **435** may attempt to decode the command into an access instruction and an address (e.g., the access instruction indicating a type of access operation to be performed at the address of the memory array **470**). In some cases, there may be an error when the memory device **410** transfers the command from the input buffer **425** and/or attempts to decode the command at decoder **435**. For example, the command may not be transferred correctly from the input buffer **425** to the decoder **435**. That is, the raw pin data (e.g., indicating the command) received at the decoder **435** may not match the data transmitted across channel **420** from the host device **405** (e.g., the data received at the input buffer **425**). In another example, the decoder **435** may incorrectly decode the command and therefore may not determine a correct access instruction and/or address.

[0079] The memory device **410** may transmit data corresponding to the access instruction (e.g., as determined based on decoding the command) from the decoder **435** to the sideband pins **496** (e.g., via a bus **480**). The sideband pins **496** may be a serial bus such that the decoder **435** sends the data serially via the sideband pins **496** and bus **497**. In some cases, the data may indicate a success (or lack thereof) of decoding the command to determine an access instruction. That is, the data may indicate an error resulting in the decoder **435** not determining a valid type of access instruction

based on the command (e.g., not successfully decoding the command) or not determining a valid address from the command. For example, the decoder **435** may transmit a logic '0' on the bus **480** in the event that an access instruction and valid address was determined from the command. Conversely, the decoder **435** may transmit a logic '1' on the bus **480** in the event that an access instruction was not determined from the command (e.g., corresponding to a failed decoding by decoder **435**). Alternatively the logic states for successfully decoding the command may be inverted. In some other cases, the data may indicate the type of access instruction determined by the decoder **435**. For example, the decoder **435** may transmit a bit (or a series of bits) to the sideband pins **496** that may map to the type of access instruction (e.g., a read instruction, a write instruction).

[0080] In the case where the decoder **435** transmits the type of access instruction to the sideband pins **496**, the host device **405** may determine if the type of access instruction determined at the decoder **435** matches a type of access instruction indicated by the command transmitted by the host device **405**. Here, the host device **405** may receive an indication of the type of access command determined at the decoder **435**. The host device **405** may compare the type of access instruction indicated via the sideband pins **496** to the type of access instruction indicated by the command transmitted by the host device **405**. By this method, the host device **405** may determine if there is an error associated with the access instruction determined at the decoder **435**.

[0081] After decoding the command received from the input buffer **425**, the decoder **435** may subsequently indicate the instruction and the address to the memory array **470** (e.g., via the bus **485**). The memory array **470** may be accessed at the indicated address and according to the indicated instruction (e.g., an instruction corresponding to a read operation, a write operation). For example, the memory array **470** may include a row decoder and a column decoder (e.g., as described with reference to FIG. 2) configured to receive an address for memory array **470** and activate a word line and select digit lines respectively based on the received address (e.g., as received from the decoder **435**). The memory array **470** may further include row access logic **465** and column access logic **475** configured to determine an accessed portion of the memory array **470** accessed during the execution of the indicated instruction. For example, as described with reference to FIG. 2, the row access logic **465** may determine a word line or a set of word lines accessed by the memory array **470**. The word line or word lines accessed may correspond to an accessed row MAT. Additionally or alternatively, the column access logic **475** may determine a digit line or a set of digit lines accessed by the memory array **470**. The digit line or digit lines accessed may correspond to an accessed column MAT. The accessed row MAT and/or the access column MAT may correspond to the accessed portion of the memory array **470**.

[0082] There may be an error corresponding to the location of the memory array **470** during the access operation. For example, the decoder **435** may incorrectly decode the command and determine an address for the access command that does not correspond to an address indicated by the command received from the host device **405**. In another example, the memory array **470** may incorrectly access a word line and/or digit line. For example, a row decoder and/or a column decoder may activate an incorrect word line or select incorrect digit lines (e.g., a different word or digit line than indicated by the address within the command received from the host **405**).

[0083] The memory device **410** may transmit (e.g., via bus **490**, sideband pins **496**, and bus **497**) information related to the accessed portion of the memory array **470** to enable the host device **405** to determine such errors. The memory device **410** may transmit an indication of the accessed portion of the memory array **470** (e.g., as determined by the row access logic **465** and/or the column access logic **475**). The indication may include an identifier for the accessed portion (e.g., a row identifier, a column identifier, a row MAT identifier, a column MAT identifier). In one example, the accessed portion may correspond to a row MAT. In some other examples, the accessed portion may correspond to a row MAT and a column MAT. In another example, the accessed portion may correspond to an intersection of a single word line and a single digit line

(e.g., a single memory cell).

[0084] The host device **405** may receive the command from the sideband pins **496**. The host device **405** may use circuitry (e.g., a LUT) in order to map the command, transmitted to the memory device **410**, to an expected portion of the memory array **470**. The expected portion may include an expected row MAT, an expected column MAT, or an expected array location (e.g., an expected column MAT and/or an expected row MAT). The host device **405** may compare the expected portion to the accessed portion indicated by the memory array **470**. The host device **405** may determine an error in the event that the expected portion does not correspond to the accessed portion indicated by the memory array **470**. Here, the memory device **410** may determine that the executed command may not have been executed properly.

[0085] The host device **405** may receive information allowing the host device **405** to determine errors such as the memory device **410** incorrectly decoding the command, the memory device **410** failing to decode the command, or the memory device **410** accessing a portion of the memory array **470** that does not match a portion of the memory array **470** indicated by the command from the host device **405**. The host device **405** may determine to retransmit the command based on receiving the information. This may allow the host device **405** to improve reliability of the memory device **410** when compared to a memory device that does not provide such error feedback. In some cases, the host device **405** may transmit a stream of commands (e.g., multiple commands that are transmitted contiguously or non-contiguously to each other) to the memory device **410**. In this example, the memory device **410** may execute the transmitted commands concurrently to transmitting information allowing the host device **405** to determine errors associated with one or more of the stream of commands. That is, normal memory operations may continue concurrently with sending information from the memory device **410** to the host device **405** that indicates decoding status for commands, decoded instructions, decoded addresses, or accessed portions of the memory array **470**. This may allow the memory device **410** to provide additional information to the host device **405** (e.g., information related to increasing reliability of the memory device **410**) without significantly decreasing read/write speeds of the memory device.

[0086] FIG. 5 illustrates an example of a process flow **500** that supports techniques for memory command verification for a memory device. The process flow **500** may implement aspects of the systems **100**, **300**, and **400**, and memory sub-array **200**. The process flow **500** may include operations performed by a host **505**, which may be an example of host devices **305** or **405** as described with reference to FIGS. 3 and 4. Host device **505** may implement aspects of the external memory controller **105** as described with reference to FIG. 1; The process flow **500** may further include operations performed by a memory device **510**, which may be an example of the memory device **110**, the memory dice **160**, or the memory sub-array **200** as described with reference to FIGS. 1 and 2, and may be examples of memory devices **310** and **410** as described with reference to FIGS. 3 and 4.

[0087] At **515**, the host device **505** may transmit a command to the memory device **510**.

[0088] At **520**, the memory device **510** may decode the command. In some cases, the memory device **510** may further generate an identifier for the command based on decoding the command. The memory device **510** may decode the command at one or more decoders.

[0089] At **525**, the memory device **510** may access a portion of the memory array of the memory device **510** in response to the command. Accessing the portion of the memory array may be based on decoding the command at **520**. The memory device **510** may further generate an identifier of the portion of the memory array accessed in response to the command. The identifier may be generated at the memory array. In some cases, the identifier may include an identifier for one or more dimensions of the memory array (e.g., identifying a row MAT, a column MAT, and/or both). In some examples, the identifier may include an indication of an MAT or sub-array of the memory array (e.g., identifying an intersection of a row MAT and a column MAT).

[0090] At **530**, the memory device **510** may optionally verify the execution of the command. Here,

the memory device **510** may decode the command received from the host device **505** at one or more verifiers. In some cases, the memory device **510** may compare the identifier of the portion of the memory array to the command to the command received from the host device **505**. In some other cases, the memory device may optionally compare the identifier for the command (e.g., as determined at **520**) to the command received from the host device **505**. In some cases, the command identifier may be generated at one or more decoders of the memory device.

[0091] At **535**, the memory device **510** may transmit signaling indicating the portion of the memory array accessed in response to the command. In some cases (e.g., in the case that the memory device **510** compares the identifier of the portion of the memory array to the command received from the host device **505** at **530**), the signaling indicating the portion of the memory array accessed in response to the command may be based on the comparison. For example, the signaling may indicate that the portion of the memory array accessed is different than a portion of the memory array indicated by the identifier of the portion of the memory array. Additionally or alternatively, the signaling may indicate that the portion of the memory array accessed is the same as a portion of the memory indicated by the identifier of the portion of the memory array. The signaling may include an indication of the MAT or sub-array of the memory array accessed in response to the command.

[0092] In some examples (e.g., in the case that the memory device **510** compares the identifier determined at one or more decoders to the command received from the host device **505** at **530**), the signaling may be based on the comparison. For example, the signaling may indicate that the identifier determined at one or more decoders corresponds to a same type of instruction as indicated by the command received from the host device **505**. In another example, the signaling may indicate that the identifier determined at one or more decoders corresponds to a different type of instruction than indicated by the command received from the host device **505**. Here, the signaling may indicate whether the command received at the memory device **510** was executed correctly or incorrectly within the memory device **510** (e.g., whether the type of command determined at the one or more decoders is the same as the type of command indicated by the command received from the host device **505**). In some instances, the signaling may be based on a comparison of decodes (e.g., a decode at a decoder and a decode at a verifier).

[0093] In some cases, the host device **505** may transmit a second command to the memory device **510**. The command and second command may be transmitted as part of a sequence (e.g., contiguously or non-contiguously). In some examples, sending the signaling at **535** may not disrupt a normal processing of the command and second command (e.g., the signaling at **535** may be done concurrently with the normal processing). The memory device **510** may transmit a second signaling to the host device **505** indicating that decoding of the second command failed to access the memory array.

[0094] FIG. **6** illustrates an example of a process flow **600** that supports techniques for memory command verification for a memory device. The process flow **600** may implement aspects of the systems **100**, **300**, and **400**, memory sub-array **200**, and process flow **500**. The process flow **600** may include operations performed by a host **605**, which may be an example of host devices **305**, **405**, or **505** as described with reference to FIGS. **3** through **5**. Host device **605** may implement aspects of the external memory controller **105** as described with reference to FIG. **1**; The process flow **600** may further include operations performed by a memory device **610**, which may be an example of the memory device **110**, the memory dice **160**, or the memory sub-array **200** as described with reference to FIGS. **1** and **2**, and may be examples of memory devices **310**, **410**, and **510** as described with reference to FIGS. **3** through **5**.

[0095] At **615**, the memory device **610** may receive a command from the host device **605**.

[0096] At **620**, the memory device may optionally decode the command.

[0097] At **625**, the memory device **610** may determine a portion of the memory that is accessed based on the command received from the host device **605**. Here, the memory device **610** may

determine a MAT or sub-array of the memory array corresponding to the portion of the memory array accessed in response to the command. The memory device **610** may determine the MAT or sub-array by determining a set of columns of the memory array corresponding to the portion of the memory array accessed in response to the command. The memory device **610** may further determine a set of rows of the memory array corresponding to the portion of the memory array accessed in response to the command. Here, the MAT or sub-array may be based on the set of columns and the set of rows. In some cases (e.g., in the case that the memory device **610** decodes the command at **620**), the memory device **610** may determine the portion of the memory that is accessed based on decoding the command.

[0098] At **630**, the memory device **610** may determine a status associated with the command based on the portion of the memory array that is accessed and the command received at the memory device. Determining the status associated with the command may be based on the MAT or the sub-array of the memory array (e.g., as determined at **625**). In some cases, the memory device **610** may determine the status by determining a location of the memory array intended to be accessed by the command sent by the host device **605**. The memory device **610** may compare the portion of the memory array accessed based on decoding the command to the location intended to be accessed by the command. The status determined at **630** may be based on the comparing the portion of the memory array accessed based on decoding the command to the location intended to be accessed by the command.

[0099] In some cases, the memory device **610** may determine the status by determining, based on the command received from the host device **605**, an instruction for the access of the memory array. The memory device **620** may further determine a type of access for the portion of the memory array accessed based at least in part on decoding the command. The status determined at **630** may be based on the comparing the type of access to the instruction.

[0100] At **635**, the memory device **610** may transmit a signaling to the host device **605** indicating the status associated with the command (e.g., as determined at **630**). In a first example, the signaling indicates a status related to the portion of the memory array accessed. In some cases, the memory device **610** may indicate an error for the command in the signaling. The error may be based on determining a mismatch between the portion of the memory array accessed in response to the command and the location intended to be accessed by the command. In some other cases, the memory device **610** may transmit a confirmation for the command in the signaling. The confirmation may be based on determining that the portion of the memory array accessed in response to the command matches the location intended to be accessed by the command.

[0101] In a second example (which may be in addition to or alternatively to the first example), the signaling indicates a status related to the type of access. In some cases, the memory device **610** may transmit an error as part of the signaling. The error may be based on determining a mismatch between the type of access and the instruction (e.g., at **630**). In some other cases, the memory device **610** may transmit a confirmation for the command as part of the signaling. The confirmation may be based on determining that the type of access matches the instruction (e.g., at **630**).

[0102] In some cases, the memory device **610** may receive a second command from the host device **605**. The memory device **610** may transmit, to the host device **605**, an indicator of a failure to decode the second command.

[0103] FIG. 7 illustrates an example of a process flow **700** that supports techniques for memory command verification for a memory device. The process flow **700** may implement aspects of the systems **100**, **300**, and **400**, memory sub-array **200**, and process flows **500** and **600**. The process flow **700** may include operations performed by a host **705**, which may be an example of host devices **305**, **405**, **505**, or **605** as described with reference to FIGS. 3 through 6. Host device **705** may implement aspects of the external memory controller **105** as described with reference to FIG. 1; The process flow **700** may further include operations performed by a memory device **710**, which may be an example of the memory device **110**, the memory array **170**, or the memory sub-array

200 as described with reference to FIGS. **1** and **2**, and may be examples of memory devices **310**, **410**, **510**, and **610** as described with reference to FIGS. **3** through **6**.

[0104] At **715**, the host device **705** may transmit, to the memory device **710**, a command to be executed on a memory array of the memory device **710**.

[0105] At **720**, the host device **705** may receive an indicator from the memory device **710**. The indicator may be of a portion of the memory array accessed by the memory device **710** in response to the command. In some cases, the indicator may further include an indicator of a type of access for the portion of the memory array accessed by the memory device **710**.

[0106] At **725**, the host device **705** may optionally determine information indicating a status of execution for the command. The status may be based on the portion of the memory array accessed in response to the command. The information may indicate a MAT or sub-array of the memory array accessed by the memory array in response to the command. In some cases, determining information indicating a status of the execution for the command may include determining whether the portion of the memory array accessed by the memory device **710** is correct for the command. In some cases, determining information indicating the status of execution for the command may include the host device **705** determining whether the type of access is correct for the command. For example, the host device **705** may determine if the type of access command indicated by the memory device **710** is the same as the type of access command indicated by the command at **715**.

[0107] At **730**, the host device **705** may process one or more additional commands for the memory device based on the indicator. In some cases, the host device **705** may process the one or more additional commands for the memory device **710** based on an indication of decoding error for a second command. Here, the host device **705** may have transmitted, to the memory device **710**, a second command to be executed on the memory array. The host device **705** may also receive, from the memory device **710**, an indicator of a decoding error for the second command.

[0108] FIG. **8** shows a block diagram **800** of a device **805** that supports memory command verification as disclosed herein. The device **805** may be an example of aspects of a memory device such as memory device **110**, memory device **310**, memory device **510**, or memory device **610** as disclosed herein with reference to FIGS. **1**, **3**, **5**, and **6**. The device **805** may include a command receiver **810**, a memory access manager **815**, a host signaling manager **820**, a command decoding component **825**, a comparison manager **830**, and a verifier component **835**. Each of these modules may communicate, directly or indirectly, with one another (e.g., via one or more buses).

[0109] The command receiver **810** may receive a command from a host device at a memory device. In some examples, the command receiver **810** may receive a second command from the host device.

[0110] The memory access manager **815** may access a portion of a memory array of the memory device in response to the command. In some examples, the memory access manager **815** may generate an identifier of the portion of the memory array accessed in response to the command. In some cases, the identifier of the portion of the memory array accessed in response to the command is generated at the memory array. In some cases, the identifier of the portion of the memory array accessed in response to the command includes an identifier for one or more dimensions of the memory array. In some cases, the identifier of the portion of the memory array accessed in response to the command includes an indication of a MAT or sub-array of the memory array.

[0111] The host signaling manager **820** may transmit signaling to the host device that indicates the portion of the memory array accessed in response to the command. In some cases, the signaling that indicates the portion of the memory array accessed in response to the command includes an indication of a MAT or sub-array of the memory array. In some examples, the signaling transmitted to the host device is based on a comparison of decodes at the one or more decoders and the one or more verifiers. In some examples, the host signaling manager **820** may transmit second signaling to the host device that indicates that decoding of the second command failed to access the memory array.

[0112] The command decoding component **825** may decode the command at the memory device, where accessing the portion of the memory array is based on decoding the command. In some examples, the command decoding component **825** may generate an identifier for the command based on decoding the command at the memory device. In some cases, the identifier for the command is generated at one or more decoders of the memory device. In some examples, the command decoding component **825** may decode the command received from the host device at one or more decoders, where accessing the portion of the memory array is based on decoding the command at the one or more decoders.

[0113] The comparison manager **830** may compare the identifier to the command received from the host device, where the signaling that indicates the portion of the memory array accessed in response to the command is based on the comparison. In some examples, the comparison manager **830** may compare the identifier for the command to the command received from the host device, where the signaling that indicates the portion of the memory array accessed in response to the command is based on the comparison.

[0114] The verifier component **835** may decode the command received from the host device at one or more verifiers.

[0115] The command receiver **810** may receive a command at a memory device. In some examples, the command receiver **810** may receive a second command from the host device at the memory device. In some examples, the command receiver **810** may receive a series of commands from the host device, where the series of commands includes the command.

[0116] The memory access manager **815** may determine a portion of a memory array that is accessed based on the command. In some examples, the memory access manager **815** may determine a MAT or a sub-array of the memory array corresponding to the portion of the memory array accessed in response to the command. In some examples, the memory access manager **815** may determine a set of columns of the memory array corresponding to the portion of the memory array accessed in response to the command. In some examples, the memory access manager **815** may determine a set of rows of the memory array corresponding to the portion of the memory array accessed in response to the command. In some examples, the memory access manager **815** may determine the MAT or the sub-array based on the set of columns and the set of rows. In some examples, the memory access manager **815** may access respective portions of the memory array in response to the series of commands, where accessing at least one of the respective portions of the memory array occurs concurrently with determining the status.

[0117] The host signaling manager **820** may transmit signaling to a host device that indicates the status associated with the command. In some examples, the host signaling manager **820** may transmit, as at least part of the signaling that indicates the status, an indication of an error for the command based on determining a mismatch between the portion of the memory array accessed in response to the command and the location intended to be accessed by the command. In some examples, the host signaling manager **820** may transmit, as at least part of the signaling that indicates the status, a confirmation for the command based on determining that the portion of the memory array accessed in response to the command matches the location intended to be accessed by the command. In some examples, the host signaling manager **820** may transmit, as at least part of the signaling that indicates the status, an error indicator for the command based on determining a mismatch between the type of access and the instruction. In some examples, the host signaling manager **820** may transmit, as at least part of the signaling that indicates the status, a confirmation for the command based on determining that the type of access matches the instruction. In some examples, the host signaling manager **820** may transmit, to the host device, an indicator of a failure to decode the second command. In some cases, the signaling transmitted to the host device includes a status that indicates whether the command received at the memory device was executed correctly or incorrectly within the memory device.

[0118] The verifier component **835** may determine a status associated with the command based on

the portion of the memory array that is accessed and the command received at the memory device. In some examples, the verifier component **835** may determine the status based on comparing the type of access to the instruction. In some examples, the verifier component **835** may determine the status associated with the command based on the MAT or the sub-array of the memory array.

[0119] The command decoding component **825** may decode the command, where the portion of the memory array is accessed based on the decoding the command. In some examples, the command decoding component **825** may determine a location of the memory array intended to be accessed by the command. In some examples, the command decoding component **825** may determine, based on the command received from the host device, an instruction for the access of the memory array. In some examples, the command decoding component **825** may determine a type of access for the portion of the memory array accessed based on decoding the command.

[0120] The comparison manager **830** may compare the portion of the memory array accessed based on decoding the command to the location intended to be accessed by the command, where determining the status is based on comparing the portion of the memory array accessed by the command to the location intended to be accessed by the command.

[0121] FIG. **9** shows a block diagram **900** of a device **905** that supports memory command verification as disclosed herein. The device **905** may be an example of aspects of a host device such as host device **405** or host device **705** as disclosed herein with reference to FIGS. **4** and **7**. The device **905** may include a command transmitter **910**, an indicator manager **915**, an additional command processor **920**, and a status manager **925**. Each of these modules may communicate, directly or indirectly, with one another (e.g., via one or more buses).

[0122] The command transmitter **910** may transmit, to a memory device, a command to be executed on a memory array of the memory device. In some examples, the command transmitter **910** may transmit, to the memory device, a second command to be executed on the memory array.

[0123] The indicator manager **915** may receive, from the memory device, an indicator of a portion of the memory array accessed by the memory device in response to the command. In some examples, the indicator manager **915** may receive, from the memory device, an indicator of a type of access for the portion of the memory array accessed by the memory device. In some examples, the indicator manager **915** may receive, from the memory device, an indicator of a decoding error for the second command.

[0124] The additional command processor **920** may process one or more additional commands for the memory device based on the indicator. In some examples, the additional command processor **920** may process the one or more additional commands for the memory device based on the indicator of the decoding error.

[0125] The status manager **925** may determine, based on the portion of the memory array accessed in response to the command, information indicating a status of execution for the command. In some cases, the information indicates a MAT or sub-array of the memory array accessed by the memory array in response to the command. In some examples, the status manager **925** may determine whether the portion of the memory array accessed by the memory device is correct for the command, where determining the information is based on determining whether the portion of the memory array accessed by the memory device is correct for the command. In some examples, the status manager **925** may determine whether the type of access is correct for the command, where determining the information is based on determining whether the type of access is correct.

[0126] FIG. **10** shows a flowchart illustrating a method **1000** that supports systems, devices, and methods for memory command verification as disclosed herein. The operations of method **1000** may be implemented by a memory device (e.g., memory device **110**, memory device **310**, memory device **510**, or memory device **610** as disclosed herein with reference to FIGS. **1**, **3**, **5**, and **6**) or its components as described herein. In some examples, a memory device may execute a set of instructions to control the functional elements of the memory device to perform the functions described herein. Additionally or alternatively, a memory device may perform aspects of the

functions described herein using special-purpose hardware.

[0127] At **1005**, the memory device may receive a command from a host device at a memory device. The operations of **1005** may be performed according to the methods described herein. In some examples, aspects of the operations of **1005** may be performed by a command receiver as described with reference to FIG. 8.

[0128] At **1010**, the memory device may access a portion of a memory array of the memory device in response to the command. The operations of **1010** may be performed according to the methods described herein. In some examples, aspects of the operations of **1010** may be performed by a memory access manager as described with reference to FIG. 8.

[0129] At **1015**, the memory device may transmit signaling to the host device that indicates the portion of the memory array accessed in response to the command. The operations of **1015** may be performed according to the methods described herein. In some examples, aspects of the operations of **1015** may be performed by a host signaling manager as described with reference to FIG. 8.

[0130] FIG. 11 shows a flowchart illustrating a method **1100** that supports systems, devices, and methods for memory command verification as disclosed herein. The operations of method **1100** may be implemented by a memory device or its components as described herein. For example, the operations of method **1100** may be performed by a memory device as described with reference to FIGS. 1, 3, 4, and 6. In some examples, a memory device may execute a set of instructions to control the functional elements of the memory device to perform the functions described herein. Additionally or alternatively, a memory device may perform aspects of the functions described herein using special-purpose hardware.

[0131] At **1105**, the memory device may receive a command from a host device at a memory device. The operations of **1105** may be performed according to the methods described herein. In some examples, aspects of the operations of **1105** may be performed by a command receiver as described with reference to FIG. 8.

[0132] At **1110**, the memory device may decode the command at the memory device, where accessing the portion of the memory array is based on decoding the command. The operations of **1110** may be performed according to the methods described herein. In some examples, aspects of the operations of **1110** may be performed by a command decoding component as described with reference to FIG. 8.

[0133] At **1115**, the memory device may access a portion of a memory array of the memory device in response to the command. The operations of **1115** may be performed according to the methods described herein. In some examples, aspects of the operations of **1115** may be performed by a memory access manager as described with reference to FIG. 8.

[0134] At **1120**, the memory device may generate an identifier of the portion of the memory array accessed in response to the command. The operations of **1120** may be performed according to the methods described herein. In some examples, aspects of the operations of **1120** may be performed by a memory access manager as described with reference to FIG. 8.

[0135] At **1125**, the memory device may compare the identifier to the command received from the host device, where the signaling that indicates the portion of the memory array accessed in response to the command is based on the comparison. The operations of **1125** may be performed according to the methods described herein. In some examples, aspects of the operations of **1125** may be performed by a comparison manager as described with reference to FIG. 8.

[0136] At **1130**, the memory device may transmit signaling to the host device that indicates the portion of the memory array accessed in response to the command. The operations of **1130** may be performed according to the methods described herein. In some examples, aspects of the operations of **1130** may be performed by a host signaling manager as described with reference to FIG. 8.

[0137] An apparatus for performing a method or methods, such as the method **1100**, is described. The apparatus may include means for receiving a command from a host device at a memory device, accessing a portion of a memory array of the memory device in response to the

command, and transmitting signaling to the host device that indicates the portion of the memory array accessed in response to the command.

[0138] In some examples, the apparatus may include features for decoding the command at the memory device, where accessing the portion of the memory array may be based on decoding the command, generating an identifier of the portion of the memory array accessed in response to the command and comparing the identifier to the command received from the host device, where the signaling that indicates the portion of the memory array accessed in response to the command may be based on the comparison.

[0139] In some cases, the identifier of the portion of the memory array accessed in response to the command may be generated at the memory array. In some instances, the identifier of the portion of the memory array accessed in response to the command includes an identifier for one or more dimensions of the memory array. In some examples, the identifier of the portion of the memory array accessed in response to the command includes an indication of a MAT or sub-array of the memory array.

[0140] In some examples, the apparatus may include features for generating an identifier for the command based on decoding the command at the memory device and comparing the identifier for the command to the command received from the host device, where the signaling that indicates the portion of the memory array accessed in response to the command may be based on the comparison.

[0141] In some cases, the identifier for the command may be generated at one or more decoders of the memory device.

[0142] In some examples, the apparatus may include features for decoding the command received from the host device at one or more decoders, where accessing the portion of the memory array may be based on decoding the command at the one or more decoders, decoding the command received from the host device at one or more verifiers, and where the signaling transmitted to the host device may be based on a comparison of decodes at the one or more decoders and the one or more verifiers.

[0143] In some cases, the signaling transmitted to the host device includes a status that indicates whether the command received at the memory device was executed correctly or incorrectly within the memory device.

[0144] In some instances, the signaling that indicates the portion of the memory array accessed in response to the command includes an indication of a MAT or sub-array of the memory array.

[0145] In some examples, the apparatus may include features for receiving a second command from the host device at the memory device and transmitting second signaling to the host device that indicates that decoding of the second command failed to access the memory array.

[0146] FIG. 12 shows a flowchart illustrating a method **1200** that supports systems, devices, and methods for memory command verification as disclosed herein. The operations of method **1200** may be implemented by a memory device or its components as described herein. For example, the operations of method **1200** may be performed by a memory device as described with reference to FIGS. 1, 3, 5, and 6. In some examples, a memory device may execute a set of instructions to control the functional elements of the memory device to perform the functions described herein. Additionally or alternatively, a memory device may perform aspects of the functions described herein using special-purpose hardware.

[0147] At **1205**, the memory device may receive a command at a memory device. The operations of **1205** may be performed according to the methods described herein. In some examples, aspects of the operations of **1205** may be performed by a command receiver as described with reference to FIG. 8.

[0148] At **1210**, the memory device may determine a portion of a memory array that is accessed based on the command. The operations of **1210** may be performed according to the methods described herein. In some examples, aspects of the operations of **1210** may be performed by a

memory access manager as described with reference to FIG. 8.

[0149] At **1215**, the memory device may determine a status associated with the command based on the portion of the memory array that is accessed and the command received at the memory device. The operations of **1215** may be performed according to the methods described herein. In some examples, aspects of the operations of **1215** may be performed by a verifier component as described with reference to FIG. 8.

[0150] At **1220**, the memory device may transmit signaling to a host device that indicates the status associated with the command. The operations of **1220** may be performed according to the methods described herein. In some examples, aspects of the operations of **1220** may be performed by a host signaling manager as described with reference to FIG. 8.

[0151] An apparatus for performing a method or methods, such as the method **1200**, is described. The apparatus may include means for receiving a command at a memory device, determining a portion of a memory array that is accessed based on the command, determining a status associated with the command based on the portion of the memory array that is accessed and the command received at the memory device, and transmitting signaling to a host device that indicates the status associated with the command.

[0152] In some cases, the apparatus may include features for decoding the command, where the portion of the memory array may be accessed based on the decoding the command.

[0153] In some examples, the apparatus may include features for determining a location of the memory array intended to be accessed by the command and comparing the portion of the memory array accessed based on decoding the command to the location intended to be accessed by the command, where determining the status may be based on comparing the portion of the memory array accessed by the command to the location intended to be accessed by the command.

[0154] In some instances, the apparatus may include features for transmitting, as at least part of the signaling that indicates the status, a confirmation for the command based on determining that the portion of the memory array accessed in response to the command matches the location intended to be accessed by the command.

[0155] In some cases, the apparatus may include features for transmitting, as at least part of the signaling that indicates the status, a confirmation for the command based on determining that the portion of the memory array accessed in response to the command matches the location intended to be accessed by the command.

[0156] In some examples, the apparatus may include features for determining, based on the command received from the host device, an instruction for the access of the memory array, determining a type of access for the portion of the memory array accessed based on decoding the command and determining the status based on comparing the type of access to the instruction.

[0157] In some instances, the apparatus may include features for transmitting, as at least part of the signaling that indicates the status, an error indicator for the command based on determining a mismatch between the type of access and the instruction.

[0158] In some cases, the apparatus may include features for transmitting, as at least part of the signaling that indicates the status, a confirmation for the command based on determining that the type of access matches the instruction.

[0159] In some examples, the apparatus may include features for receiving a second command from the host device and transmitting, to the host device, an indicator of a failure to decode the second command.

[0160] In some instances, the apparatus may include features for determining a MAT or a sub-array of the memory array corresponding to the portion of the memory array accessed in response to the command and determining the status associated with the command based on the MAT or the sub-array of the memory array.

[0161] In some cases, the apparatus may include features for determining a set of columns of the memory array corresponding to the portion of the memory array accessed in response to the

command, determining a set of rows of the memory array corresponding to the portion of the memory array accessed in response to the command and determining the MAT or the sub-array based on the set of columns and the set of rows.

[0162] In some examples, the apparatus may include features for receiving a series of commands from the host device, where the series of commands includes the command and accessing respective portions of the memory array in response to the series of commands, where accessing at least one of the respective portions of the memory array occurs concurrently with determining the status.

[0163] FIG. 13 shows a flowchart illustrating a method **1300** that supports systems, devices, and methods for memory command verification as disclosed herein. The operations of method **1300** may be implemented by a host device or its components as described herein. For example, the operations of method **1300** may be performed by a host device as described with reference to FIGS. 4 and 7. In some examples, a host device may execute a set of instructions to control the functional elements of the host device to perform the functions described herein. Additionally or alternatively, a host device may perform aspects of the functions described herein using special-purpose hardware.

[0164] At **1305**, the host device may transmit, to a memory device, a command to be executed on a memory array of the memory device. The operations of **1305** may be performed according to the methods described herein. In some examples, aspects of the operations of **1305** may be performed by a command transmitter as described with reference to FIG. 9.

[0165] At **1310**, the host device may receive, from the memory device, an indicator of a portion of the memory array accessed by the memory device in response to the command. The operations of **1310** may be performed according to the methods described herein. In some examples, aspects of the operations of **1310** may be performed by an indicator manager as described with reference to FIG. 9.

[0166] At **1315**, the host device may process one or more additional commands for the memory device based on the indicator. The operations of **1315** may be performed according to the methods described herein. In some examples, aspects of the operations of **1315** may be performed by an additional command processor as described with reference to FIG. 9.

[0167] An apparatus for performing a method or methods, such as the method **1300**, is described. The apparatus may include means for transmitting, to a memory device, a command to be executed on a memory array of the memory device, receiving, from the memory device, an indicator of a portion of the memory array accessed by the memory device in response to the command, and processing one or more additional commands for the memory device based on the indicator.

[0168] In some examples, the apparatus may include features for determining, based on the portion of the memory array accessed in response to the command, information indicating a status of execution for the command. In some cases, the information indicates a MAT or sub-array of the memory array accessed by the memory array in response to the command.

[0169] In some instances the apparatus may include features for determining whether the portion of the memory array accessed by the memory device may be correct for the command, where determining the information may be based on determining whether the portion of the memory array accessed by the memory device may be correct for the command.

[0170] In some cases, the apparatus may include features for receiving, from the memory device, an indicator of a type of access for the portion of the memory array accessed by the memory device and determining whether the type of access may be correct for the command, where determining the information may be based on determining whether the type of access may be correct.

[0171] In some examples, the apparatus may include features for transmitting, to the memory device, a second command to be executed on the memory array, receiving, from the memory device, an indicator of a decoding error for the second command and processing the one or more additional commands for the memory device based on the indicator of the decoding error.

[0172] It should be noted that the methods described above describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Further, aspects from two or more of the methods may be combined. [0173] In some examples, an apparatus for memory command verification may perform aspects of the functions described herein using general- or special-purpose hardware. The apparatus may include an interface coupled with a host device and operable to receive a command from the host device, a memory array coupled with the interface and operable to receive the command and determine a portion of the memory array that is accessed based on the command, and circuitry coupled with the memory array and operable to transmit, to the host device, signaling that indicates the portion of the memory array accessed based on the command.

[0174] In some cases, the signaling that indicates the portion of the memory array accessed based on the command includes an indication of a MAT or sub-array of the memory array.

[0175] In some examples, the signaling that indicates the portion of the memory array accessed based on the command includes an identifier for one or more dimensions of the memory array.

[0176] In some other examples, an apparatus for memory command verification may perform aspects of the functions described herein using general- or special-purpose hardware. The apparatus may include an interface coupled with a host device and operable to receive a command from the host device, a decoder coupled with the interface and operable to obtain a decoded command for memory access based on the command, a memory array coupled with the decoder and operable to receive the decoded command and determine a portion of the memory array accessed by the decoded command, a verifier coupled with the interface and the memory array and operable to, receiving the command from the interface, determining a status associated with the memory access based on the indicator of the portion of the memory array accessed by the command and the command, and transmitting an indicator of the status to the host device.

[0177] In some cases, the verifier may be operable to determine a location of the memory array intended to be accessed by the command and determine the status based on comparing the portion of the memory array accessed by the decoded command to the location intended to be accessed by the command.

[0178] In some instances, the verifier may be operable to determine an instruction for the access of the memory array according to the command, receive, from the memory array, a type of access for the portion of the memory array accessed in response to the decoded command and determine the status based on comparing the type of access to the instruction.

[0179] In some examples, the decoder may be further operable to determine a failure to decode a second command, the second command received by the interface from the host device and the verifier may be further operable to transmit, to the host device, an indicator of the failure to decode the second command.

[0180] In some cases, the memory array may be further operable to determine a MAT or a sub-array of the memory array corresponding to the portion of the memory array accessed in response to the command and the verifier may be further operable to determine the status associated with the command based on the MAT or the sub-array of the memory array.

[0181] Although certain features may be described herein with respect to or in the context of DRAM technology, this is for illustrative purposes only, and one of ordinary skill in the art will appreciate that the teachings herein may be applied to any type of memory device. For example, the teachings herein may be applied to volatile or non-volatile memory devices such as magnetic hard disks, random access memory (RAM), read-only memory (ROM), dynamic RAM (DRAM), synchronous dynamic RAM (SDRAM), ferroelectric RAM (FeRAM), magnetic RAM (MRAM), resistive RAM (RRAM), flash memory, phase change memory (PCM), and others.

[0182] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be

represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof. Some drawings may illustrate signals as a single signal; however, it will be understood by a person of ordinary skill in the art that the signal may represent a bus of signals, where the bus may have a variety of bit widths.

[0183] As used herein, the term “virtual ground” refers to a node of an electrical circuit that is held at a voltage of approximately zero volts (0V) but that is not directly coupled with ground.

Accordingly, the voltage of a virtual ground may temporarily fluctuate and return to approximately 0V at steady state. A virtual ground may be implemented using various electronic circuit elements, such as a voltage divider consisting of operational amplifiers and resistors. Other implementations are also possible. “Virtual grounding” or “virtually grounded” means connected to approximately 0V.

[0184] The terms “electronic communication,” “conductive contact,” “connected,” and “coupled” may refer to a relationship between components that supports the flow of signals between the components. Components are considered in electronic communication with (or in conductive contact with or connected with or coupled with) one another if there is any conductive path between the components that can, at any time, support the flow of signals between the components. At any given time, the conductive path between components that are in electronic communication with each other (or in conductive contact with or connected with or coupled with) may be an open circuit or a closed circuit based on the operation of the device that includes the connected components. The conductive path between connected components may be a direct conductive path between the components or the conductive path between connected components may be an indirect conductive path that may include intermediate components, such as switches, transistors, or other components. In some cases, the flow of signals between the connected components may be interrupted for a time, for example, using one or more intermediate components such as switches or transistors.

[0185] The term “coupling” refers to condition of moving from an open-circuit relationship between components in which signals are not presently capable of being communicated between the components over a conductive path to a closed-circuit relationship between components in which signals are capable of being communicated between components over the conductive path. When a component, such as a controller, couples other components together, the component initiates a change that allows signals to flow between the other components over a conductive path that previously did not permit signals to flow.

[0186] The term “isolated” refers to a relationship between components in which signals are not presently capable of flowing between the components. Components are isolated from each other if there is an open circuit between them. For example, two components separated by a switch that is positioned between the components are isolated from each other when the switch is open. When a controller isolates two components, the controller affects a change that prevents signals from flowing between the components using a conductive path that previously permitted signals to flow.

[0187] The devices discussed herein, including a memory array, may be formed on a semiconductor substrate, such as silicon, germanium, silicon-germanium alloy, gallium arsenide, gallium nitride, etc. In some cases, the substrate is a semiconductor wafer. In other cases, the substrate may be a silicon-on-insulator (SOI) substrate, such as silicon-on-glass (SOG) or silicon-on-sapphire (SOP), or epitaxial layers of semiconductor materials on another substrate. The conductivity of the substrate, or sub-regions of the substrate, may be controlled through doping using various chemical species including, but not limited to, phosphorous, boron, or arsenic. Doping may be performed during the initial formation or growth of the substrate, by ion-implantation, or by any other doping means.

[0188] A switching component or a transistor discussed herein may represent a field-effect transistor (FET) and comprise a three terminal device including a source, drain, and gate. The terminals may be connected to other electronic elements through conductive materials, e.g., metals.

The source and drain may be conductive and may comprise a heavily-doped, e.g., degenerate, semiconductor region. The source and drain may be separated by a lightly-doped semiconductor region or channel. If the channel is n-type (i.e., majority carriers are electrons), then the FET may be referred to as a n-type FET. If the channel is p-type (i.e., majority carriers are holes), then the FET may be referred to as a p-type FET. The channel may be capped by an insulating gate oxide. The channel conductivity may be controlled by applying a voltage to the gate. For example, applying a positive voltage or negative voltage to an n-type FET or a p-type FET, respectively, may result in the channel becoming conductive. A transistor may be “on” or “activated” when a voltage greater than or equal to the transistor's threshold voltage is applied to the transistor gate. The transistor may be “off” or “deactivated” when a voltage less than the transistor's threshold voltage is applied to the transistor gate.

[0189] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details to providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form to avoid obscuring the concepts of the described examples.

[0190] In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

[0191] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0192] The various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

[0193] The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described above can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations. Also, as used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the

phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

[0194] The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein, but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A method, comprising: receiving a first command and a second command at a memory device; accessing a first location of a memory array of the memory device in response to the first command; and transmitting signaling to a host device associated with one or more errors that are detected in the second command, wherein accessing the first location occurs at least partially concurrently with transmitting the signaling to the host device.
2. The method of claim 1, further comprising: accessing a second location of the memory array in response to the second command.
3. The method of claim 1, further comprising: decoding the first command, wherein the first location is accessed based at least in part on decoding the first command.
4. The method of claim 1, wherein transmitting the signaling comprises: setting a voltage of a pin to indicate the one or more errors.
5. The method of claim 1, further comprising: determining, based at least in part on the first command, an instruction for access of the memory array; and determining one or more second errors based at least in part on comparing a type of access for the first location to the instruction.
6. The method of claim 1, further comprising: determining a matrix (MAT) or a sub-array of the memory array corresponding to the first location; and determining one or more second errors associated with the first command based at least in part on the MAT or the sub-array of the memory array.
7. The method of claim 1, wherein the memory device comprises a dynamic random access memory (DRAM), and wherein the first command and the second command are received at the DRAM.
8. A method, comprising: receiving a first command and a second command at a memory device; accessing a first location of memory array of the memory device in response to the first command, wherein accessing the first location occurs at least partially concurrently with detecting one or more errors in the second command; and transmitting signaling to a host device that indicates the one or more errors detected in the second command.
9. The method of claim 8, further comprising: accessing a second location of the memory array in response to the second command.
10. The method of claim 8, further comprising: decoding the first command, wherein the first location is accessed based at least in part on decoding the first command.
11. The method of claim 8, wherein transmitting the signaling comprises: setting a voltage of a pin to indicate the one or more errors.
12. The method of claim 8, further comprising: determining, based at least in part on the first command, an instruction for access of the memory array; and determining one or more second errors based at least in part on comparing a type of access for the first location to the instruction.
13. The method of claim 8, further comprising: determining a matrix (MAT) or a sub-array of the

memory array corresponding to the first location; and determining one or more second errors associated with the first command based at least in part on the MAT or the sub-array of the memory array.

14. The method of claim 8, wherein the memory device comprises a dynamic random access memory (DRAM), and wherein the first command and the second command are received at the DRAM.

15. A memory device, comprising: a memory array; and processing circuitry coupled with the memory array and configured to cause the memory device to: receive a first command and a second command at the memory device; access a first location of the memory array in response to the first command; and transmit signaling to a host device associated with one or more errors that are detected in the second command, wherein accessing the first location occurs at least partially concurrently with transmitting the signaling to the host device.

16. The memory device of claim 15, wherein the processing circuitry are further configured to cause the memory device to: access a second location of the memory array in response to the second command.

17. The memory device of claim 15, wherein the processing circuitry are further configured to cause the memory device to: decode the first command, wherein the first location is accessed based at least in part on decoding the first command.

18. The memory device of claim 15, wherein, to transmit the signaling, the processing circuitry are configured to cause the memory device to: set a voltage of a pin to indicate the one or more errors.

19. The memory device of claim 15, wherein the processing circuitry are further configured to cause the memory device to: determine, based at least in part on the first command, an instruction for access of the memory array; and determine one or more second errors based at least in part on comparing a type of access for the first location to the instruction.

20. The memory device of claim 15, wherein the memory device comprises a dynamic random access memory (DRAM), and wherein the first command and the second command are received at the DRAM.
